

EARLY CAMBRIAN SHOSHONITIC MAGMATISM AT THE NORTHEAST OF THE SIBERIAN CRATON, RUSSIA

Demonterova E.I., Lomyga V.P., Ivanov A.V.

Institute of the Earth's Crust SB RAS, Irkutsk, Russia, dem@crust.irk.ru

Early Cambrian magmatic events mark one of the most intriguing periods of Earth history. For example, at this time, the hyperactivity of magnetic field was recorded (Pavlov et al., 2018) and marine invertebrates explosively diversified (Bowring et al., 1993). The factual basis for the investigation of the Early Cambrian events is the sedimentary and igneous formations of the north-eastern Siberia, where Verkhoyansky orogenic belt joins the Siberian Craton units (Bowring et al., 1993; Khudoley et al., 2013; Prokopiev et al., 2016; Pavlov et al., 2018). This region is remote and difficult for access and due to this it may keep some unique information, which cost to be revealed. Here we report new data on chemical and isotopic compositions of mafic dykes and sills of the Chekurovka anticline, which was not recorded by previous studies (Khudoley et al., 2013; Prokopiev et al., 2016).

The studied mafic rocks were collected during 2017 field campaign along Neleger (right tributary of Lena). Dykes and sills are located within Neoproterozoic sediments of the Neleger and Kharayutekh Formations and Early Cambrian Tyuser Formation (Oleinikov, 1983). At the base of the Tyuser Formation conglomerates are located whose pebbles have rhyolitic composition. Zircons from these pebbles were dated by U-Pb method using SHRIMP independently by (Bowring et al., 1993) and (Prokopiev et al., 2016) and the resulting age estimations 534.6 ± 0.4 Ma and 532.6 ± 3.1 Ma, respectively, were taken as the age of the Pre-Cambrian – Cambrian boundary. In previous studies (Khudoley et al., 2013; Prokopiev et al., 2016), mafic igneous rocks were analyzed mainly from Neoproterozoic part of the section. Only few Early Cambrian samples were also analyzed. In our study, we analyzed only Early Cambrian rocks, extending the available information into a younger period.

Analytical work was done using equipment belonging to the Centre for Geodynamics and Geochronology, Institute of the Earth's Crust SB RAS (Irkutsk, Russia). Major and trace element concentrations of the samples have been determined by XRF and ICP-MS, respectively, using lithium tetraborate and metaborate fusion for sample preparation (Pashkova et al., 2019). The $^{87}\text{Sr}/^{86}\text{Sr}$, $^{143}\text{Nd}/^{144}\text{Nd}$, $^{87}\text{Rb}/^{86}\text{Sr}$, and $^{147}\text{Sm}/^{144}\text{Nd}$ isotopic ratios were obtained by isotope dilution TIMS using a Finnigan MAT 262 mass-spectrometer according a procedure described in (Ivanov et al., 2018).

In TAS diagram (Le Bas, Streckeisen, 1991), mafic rocks of the Chekurovka anticline fall into the fields of basalt, trachybasalt, tephrite and picritic basalts. SiO_2 concentrations vary from 41 to 51 wt.% at $\text{Na}_2\text{O}+\text{K}_2\text{O}$ between 3 and 5.9 wt.%. All rocks belong to high-Ti rock series with TiO_2 between 2.4 and 6.5 wt. %. Unlike to the previous studies (Khudoley et al., 2013; Prokopiev et al., 2016), some of our samples are high in K_2O and classified as absarokites, thus mafic end-member of the shoshonitic series rocks (Fig. 1). $\text{K}_2\text{O}/\text{Na}_2\text{O}$ ratio in these rocks varies between 4.4 and 18. It is interesting, that rhyolites from conglomerate pebbles are also characterized by potassic affinity and can be classified as comendites-pantellerites (Prokopiev et al., 2016). Our study shows, that there was also mafic volcanism of the shoshonitic series in the region.

In primitive mantle-normalized diagrams, most common basaltic rocks are characterized by enriched spectra, typical for intraplate tectonic settings. This was taken as evidence for their origin because of rifting between South-Taimyr and West-Verkhoyan passive continental margins (Khudoley et al., 2013; Prokopiev et al., 2016). Absarokites are characterized by more differentiated trace-element spectra with several peaks and troughs in primitive mantle-normalized diagrams. Irrespective basaltic or shoshonitic type rocks, they show positive ϵNd_t values (+3 – +9). ϵSr_t values vary from -8 to + 67, with absarokites shifted towards the higher values. Considering that the same absarokites are characterized by high Sr concentrations (650-2100 ppm), the Sr isotopic values for them were likely not modified by secondary processes.

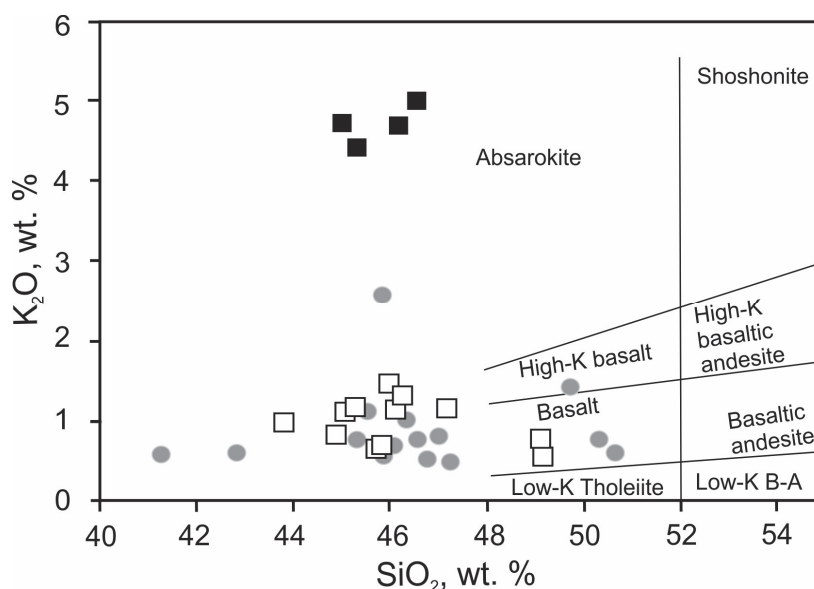


Figure 1. SiO₂-K₂O diagram for Neoproterozoic – Early Cambrian mafic rocks of the northeastern Siberia. Grey circles are from (Prokopiev et al., 2016). White and black squares are new authors' data.

Shoshonite origin is debated, though majority considers this rock type as originated in orogenic setting. If so, the appearance of comendites and pantellerites at the Pre-Cambrian – Cambrian boundary and absarokites further in time in Early Cambrian marks gradual change of the tectonic setting from rifting to compression. The significance of the recorded change in the igneous rock compositions awaits its understanding in terms of tectonic reconstructions.

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THE ZHIDOY MASSIF OF ALKALINE ROCKS AND CARBONATITES AND ITS ORE POTENCIAL (IRKUTSK REGION)

Dokuchits E.Yu., Alymova N.V., Vladykin N.V.

Institute of Geochemistry SB RAS, Irkutsk, Russia, esfor@rambler.ru; alymova@igc.irk.ru; vlad@igc.irk.ru

Abstract. In this work geological structure of the Zhidoi massif and its ore potencial has been described. The scheme of the massif magmatism has been constructed. Double correlation plots of petrogenic elements of rocks of the massif in which the unified trend of rock structures is observed, are given for verification of correctness of the scheme of magmatism. Spectra of TR and spider diagrams of concentrations of rare elements in rocks of the massif are given. Pyroxenites, early rocks of the massif are ores on titanium. Titanium concentrates in three minerals: titanomagnetite, ilmenite and perovskite. The main type of titanium ores is perovskitic type, it is known only in the Zhidoi massif. Mantle sources of primary magma of the massif are concluded on the basis of geochemistry of isotopes of Sr and Nd.

Keywords: alkaline massif, carbonatites, titanium ores, perovskite, mantle sources.

Composition and location. At the present time two important objects in Eastern Sayan mountains of the Irkutsk region are preparing for mining on strategic rare elements: the Zashikhinsky massif (on Nb, Ta, Zr) (Alymova, Vladykin, 2018), and the Zhidoy massif (on Ti). Titanium is a strategic mineral, which is necessary in industry. Titanium concentrate were previously provided to Russia from gravel deposits of Ukraine, but now this element is in short supply. In Russia titanium deposits are presented by titanomagnetite, and it is difficult to extract titanium from it. In Irkutsk region titanium deposits were discovered in recent years in the Murun massif, where they are represented by quartz zones, pure TiO_2 - brookite, anatase and rutile (Vladykin, 2016; Dokuchits, Vladykin, 2018), and in the Zhidoy massif, where titanium deposits are represented by pyroxenite with perovskite, ilmenite and titanomagnetite. The Murun massif is located in inaccessible area of the Irkutsk region while the Zhidoy massif is located in 80 km from Irkutsk, close to Talyany town. Ores of the Zhidoy massif are magmatic pyroxenites of the main phase, occupying 95% of area of the massif with outcrops.

There are four types of pyroxenites with different varieties of successive Ti-ores:

- 1) pyroxenite with titanomagnetite (broken down to magnetite and ilmenite) without perovskite;
- 2) pyroxenite with titanomagnetite (broken down to magnetite and ilmenite) with separate segregations of ilmenite;
- 3) pyroxenite with titanomagnetite and ilmenite enriched in perovskite in 7 %;
- 4) pyroxenite enriched in titanite with little titanomagnetite and without perovskite (non ore-bearing).

The massif is not well researched. Beyond pyroxenites there are outcrops of veins bodies of ijolites, alkaline and nepheline syenites. Previously only two small bedrock outcrops (up to one meter) of carbonatite dykes were known. At the present time by the performed drilling operations which were made, a carbonatite body with thickness 60 m was found. In one of its sides a rare-metal zircon-pyrochlore mineralization has occurred. It's similar with those of the Belaya Zima carbonatite province, to which the Zhidoy massif belongs. That's why carbonatites can be of industrial interest as sources of rare elements (Alymova, Vladykin, 2018). In some publications this massif called Zadoy (Rasskazov et al., 2007), which leads to confusion, because there is another Zadoy massif of ultrabasic rocks in Prisayanye area, that's why it's necessary to assign the old name Zhidoy to the massif.

The Zhidoy massif is located in 80 km from Irkutsk, in the Nephelinovy Creek when flowing into the Bolshoy Zhidoy river, in early-precambrian Sharyzhalsky block of the Prisayanye basement high of the Siberian platform. It cuts through biotite-hornblend gneisses. Phenites are originated in north part of the massif in exocontact part, they were explored by boreholes, drilled in contact wallrocks. The massif covers an area of 0,85 km² and its territory is horseshoe-shaped, with the length of intrusion of 2 km and width of 600 m. According to a geophysical data rocks of the massif continue all the way to the Bolshoy Zhidoy river.

The massif of the alkaline-ultrabasic carbonatite formation contains 90% of ore pyroxenites with different compositions of titanomagnetite, ilmenite and perovskite. It is considered that the massif has four phases of intrusion: pyroxenites, ijolites, syenites and calcite carbonatites (Konev, 1917; Rasskazov et al., 2007). Last three phases form small dykes and can be recognized as dykes of additional intrusions. The massif is hypabyssal, as evidenced by fine-grained pyroxenites; also it's possible that there are ore-bearing carbonatites containing rare elements in deeper layers. In some places carbonatites have silicate-carbonate composition, including pure albite. That's why one of the main challenges is detailed research of these carbonatites on potential to rare-metal productivity. Data on isotopic research on Sr and Nd in five varieties of rocks had been received in collaborative work with geochemists from the Shinsu University and Niigara University, Japan; according to these data the age of the massif is 570-610 mln years (Morikiyo et al., 2000; Vladykin et al., 2005). It's necessary to continue the study of isotope geochemistry and clarify the age of the massif by zircons using the U-Zr method of analysis.

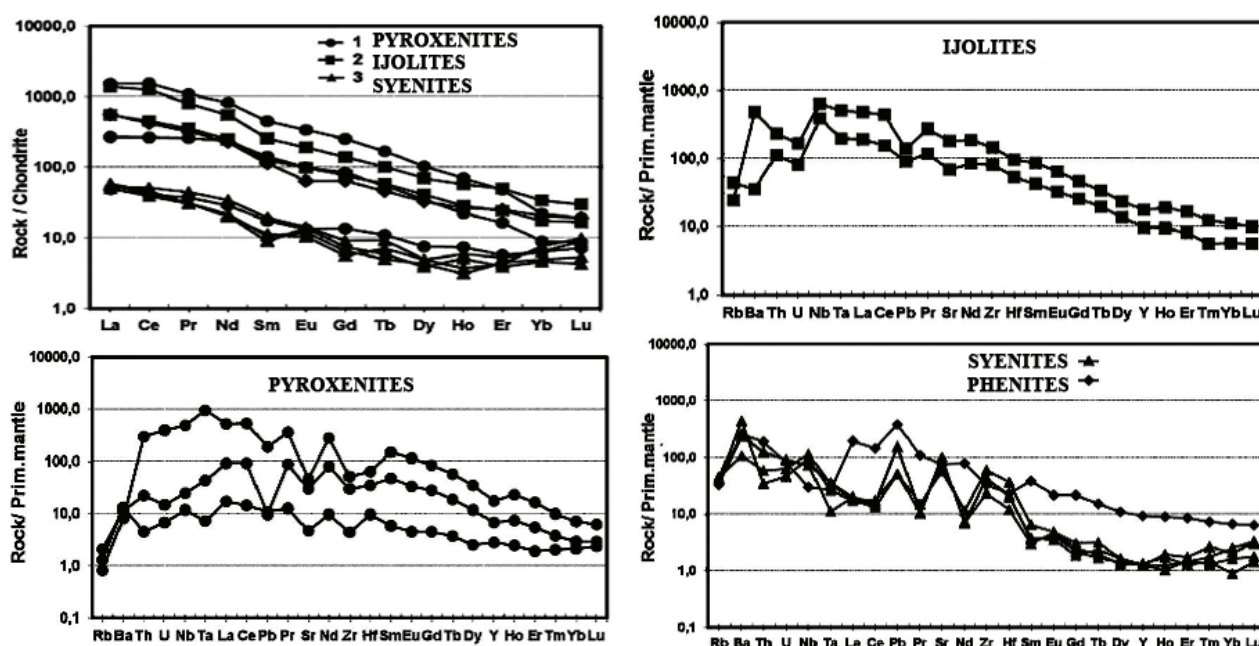


Figure. Spectra of REE and spider plots of rare elements in rocks of the Zhidoy massif.

Geochemical research. Studies of concentration of rare elements in rocks by the ICP-MS method had been made (IGC SB RAS, analyst - Mitrofanova A.Yu.). Spectra and spider plots of rare elements are shown in figure. As shown in plots, spectra of REE of rocks of the massif are identical, reflecting the genetic connection of these rocks. The highest concentrations are observed in perovskite pyroxenites and the lowest are in syenites. Increasing concentrations of high lanthanides are observed in syenites starting with Ho. Behavior of rare elements in pyroxenites is similar to those of ijolites. Negative anomalies of Pb, Sr, Zr and Hf and flat curves of REE are observed.

Different trends are observed in syenites: distinctly positive anomalies of Sr, Pb, Zr, Hf, which can be explained by differentiation processes in syenites, which are late rocks. A distribution of elements in phenites is different from those of magmatic rocks.

Conclusions:

1. Three varieties of ore-bearing pyroxenites with titanomagnetite, ilmenite and perovskite are given.
2. Common trends of compositions of rock-forming elements are observed in petrochemical diagrams reflecting the genetic connection of these rocks and their crystallization from primary parent magma.
3. Geochemical data are also confirm genetic connection of rocks of the Zhidoy massif.

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